

# Wages and Employment in a Single Labour Market

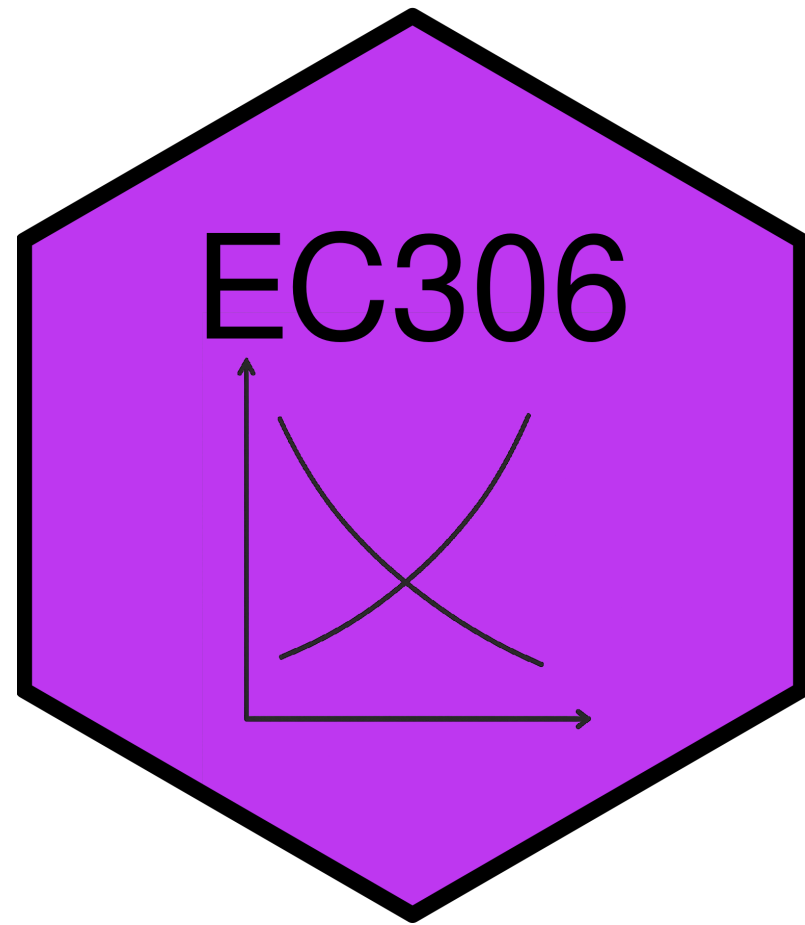
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EC306- Wages and Employment

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# Goals of This Section

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- Bring together demand and supply to determine wages and employment
- Discuss the effect of market power in goods market on equilibrium wages and employment
- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Analyze minimum wages

# Introduction

# Introduction

- We have analyzed labour demand and supply separately in detail
- The equilibrium wage and employment is determined by the interaction of the two
- Market power plays an important role
  - In both goods markets and labour markets
- With a full understanding on how wages and employment are determined, we can analyze policies
- The minimum wage is one of the most prominent labour market policies
  - In theory, it leads to unemployment
  - The evidence is more mixed

# Competitive Firms

# Introduction

- We start with firms that are competitive in the goods and labour markets
- Perfect competition in both implies that
  - Firms take the market wage as given
  - They take the market price as given
  - Competition means they are too small to influence either
- We analyze the equilibrium wage and employment for a homogeneous type of labour
  - All workers are the same

# Market Demand and Supply

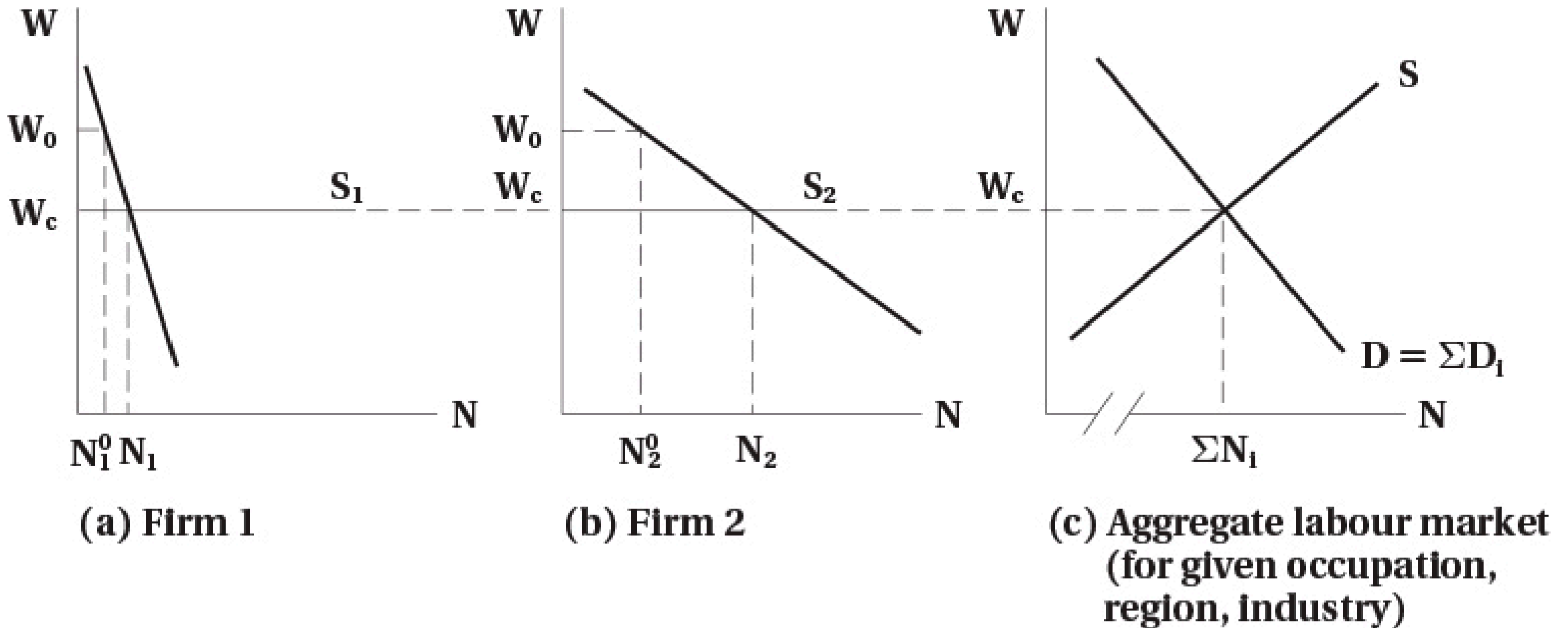
- The market demand and supply are the sum of the individual firms' demand and supply at each wage
- Here we determine the demand and supply curve for a single labour market
  - Could be an specific occupation in a specific region
  - Example: electricians in Ontario, professors in Canada, etc.
  - How the market is defined depends on context
  - But it is for a single type of labour



# Market Demand and Supply

- Example is two firms with different elasticities of demand
  - Each firm demand curve is their value of marginal product
  - Firms may face different product prices if they sell different things (but hire the same labour)
- Each firm faces the same wage in the labour market
  - They perceive the supply curve as horizontal in the long run
  - They take the wage as given and can get as much labour as they want at that wage
  - Market supply is upward sloping
- Market demand is horizontal sum of individual demands
  - Add up the quantities demanded at each wage for each firm
  - Repeat for all wages

# Market Demand and Supply

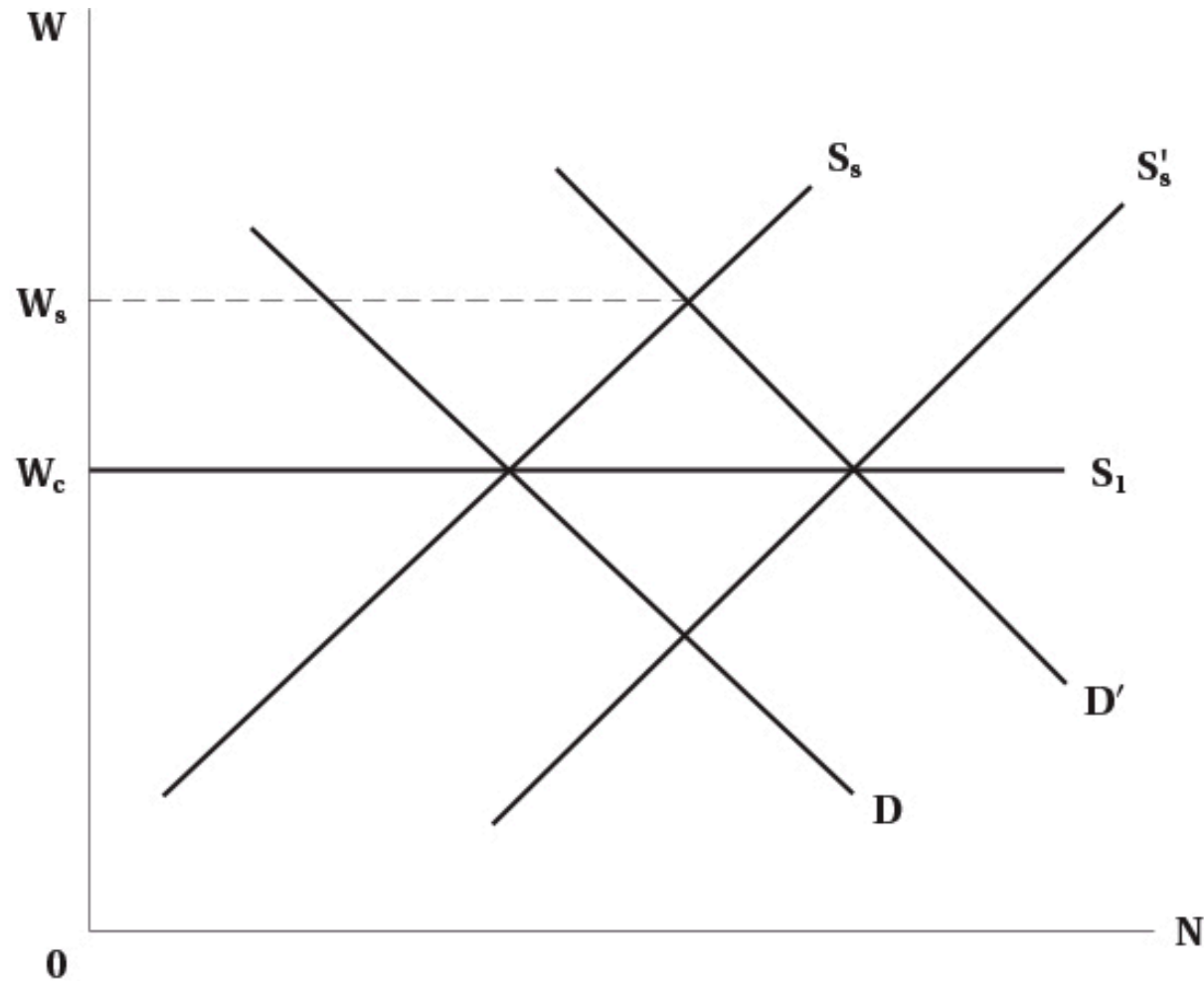


**FIGURE 7.1 Competitive Product and Labour Markets**

# Firm Demand and Supply in the Short Run

- Above we analyzed the longer run
  - Firms can get as much labour as they want at the going wage
- In the short run, firms may face an upward sloping supply curve
  - They have to pay higher wages to attract more workers
  - This could be because of contracts, or other frictions
- If demand rises, they would pay higher wages to attract workers
- But because wages are high, over the long term more workers enter the market
  - Short run supply shifts out
  - Wage falls back down to long run level

# Firm Demand and Supply in the Short Run



**FIGURE 7.2** Short-Run and Long-Run Labour Supply Schedules to the Firm

# Imperfect Competition in the Product Market

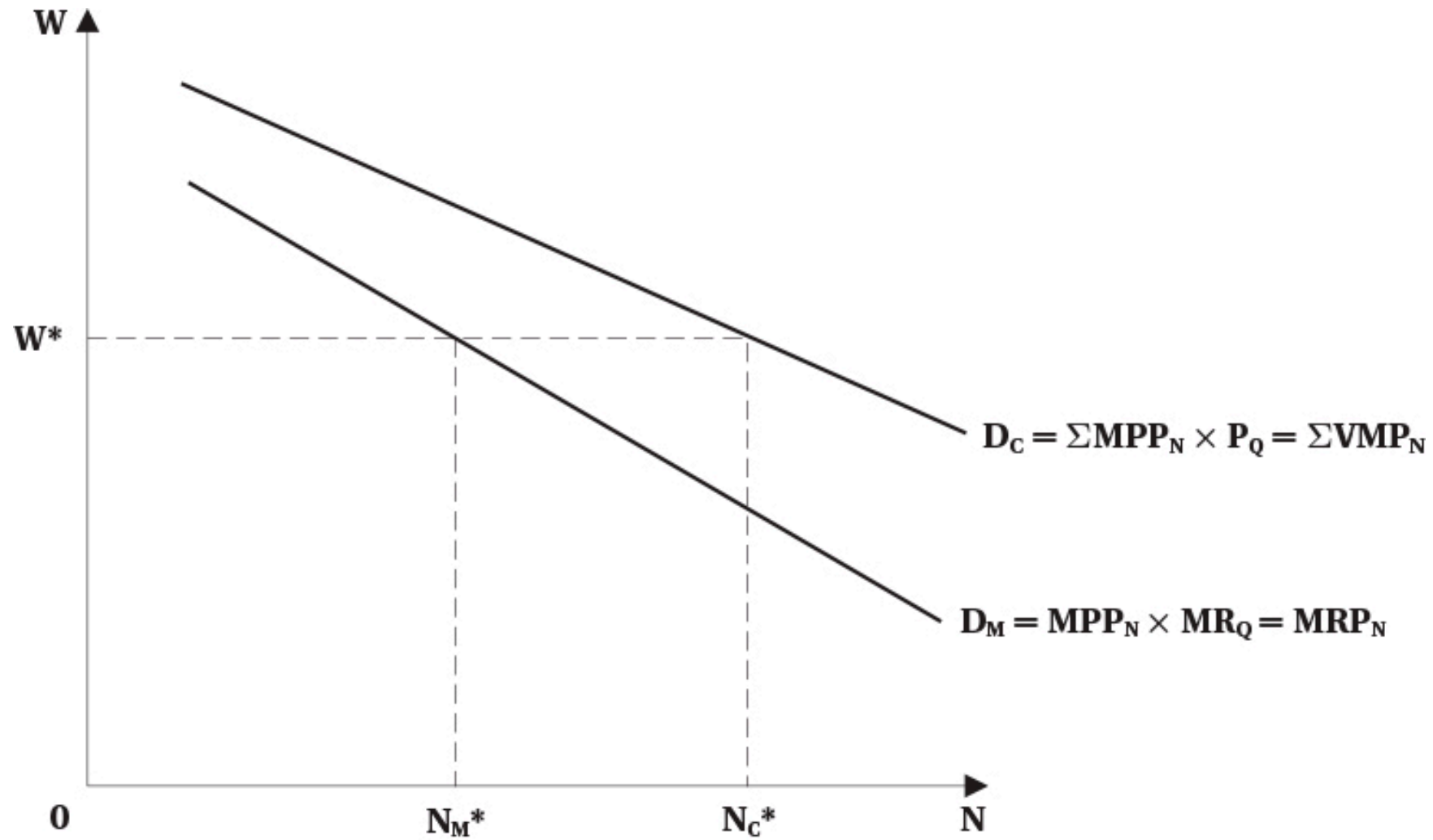
# Monopoly

- In competitive markets there are many sellers of a product
- In a monopoly, there is only one
- This gives the firm market power
  - The ability to set prices above marginal cost
- For a monopolist, the demand curve is where wage equals marginal revenue product

$$w = MR \times MP P_N$$

- For a monopolist, marginal revenue is less than price
  - Because to sell more, they have to lower price on all units sold
  - So marginal revenue product is less than value of marginal product ( $VMP$ )

# Monopoly



**FIGURE 7.3 Monopolist versus Competitive Demand for Labour**

# Monopoly

- Because monopolists have market power
  - They can raise the price of their product above competitive level
  - This will decrease output
  - They demand less labour
- Monopolists therefore will hire fewer workers
  - Compared to a competitive firm in the same market
- Note that power in the product market does not mean power in the labour market
  - The firm is still one of many hiring workers in a particular labour market



# Monopolistic Competition and Oligopoly

- Many markets are neither perfectly competitive nor pure monopoly
- Monopolistic competition
  - Many firms selling differentiated products
  - Each firm has some market power because of product differentiation
  - Can set prices above marginal cost
- Oligopoly
  - A few firms dominate the market
  - Each firm has significant market power
  - But must consider reactions of competitors when setting prices
- Demand for labour is still based on marginal revenue product
  - Which is less than value of marginal product
  - So firms hire fewer workers than in competitive markets



# Additional Details

- All of the previous analysis assumes that labour markets are perfectly competitive
  - Firms can hire all labour they want at the given wage
- That is not always the case in reality when firms have power in the product market
  - Employees may unionize
  - Firms may hire a large number of the market's workers
- In these cases, firms may not face a perfectly elastic supply curve for labour
- Later we will analyze imperfect competition in the labour market

# Math of Labour Market Equilibrium

# Introduction

- We know that market equilibrium is where demand and supply are equal
- You can do that graphically or mathematically
- In this section we cover both
- To make the math easier, we use linear demand and supply curves
  - This is just for simplicity
  - The same principles apply with non-linear curves

# Demand and Supply Functions

- The demand and supply for labour are functions of wages and other factors
  - Demand:  $N^D = f(w, Z)$ 
    - $Z$  is a list of other factors affecting demand
  - Supply:  $N^S = g(w, X)$ 
    - $X$  is a list of other factors affecting supply
- $Z$  and  $X$  are things that shift the curves
  - For example, technology, prices of other inputs, preferences, demographics, etc.
  - We call these **exogenous** variables
    - Determined outside the model
- $N^D$ ,  $N^S$ , and  $w$  are **endogenous**
  - Determined within the model

# Demand and Supply Functions

- Assume the demand and supply functions are linear
  - Demand:  $N^D = a + bw$ 
    - $a, b$  are constants
  - Supply:  $N^S = c + fw$ 
    - $c, f$  are constants
- To find equilibrium, set demand equal to supply

$$a + bw^* = c + fw^*$$

# Demand and Supply Functions

- Rearranging to solve for equilibrium wage:

$$w^* = \frac{c - a}{b - f}$$

- Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af}{b - f}$$

- Use these equations to analyze how changes in exogenous variables

- Changing  $a$  would shift the demand curve

- Example: increase in product price

- Changing  $c$  would shift the supply curve

- Example: increase in population



# Demand and Supply Functions

- To draw the demand and supply curves, we need  $w$  on the left side
- Rearranging the demand function:

$$w = \frac{N^D - a}{b}$$

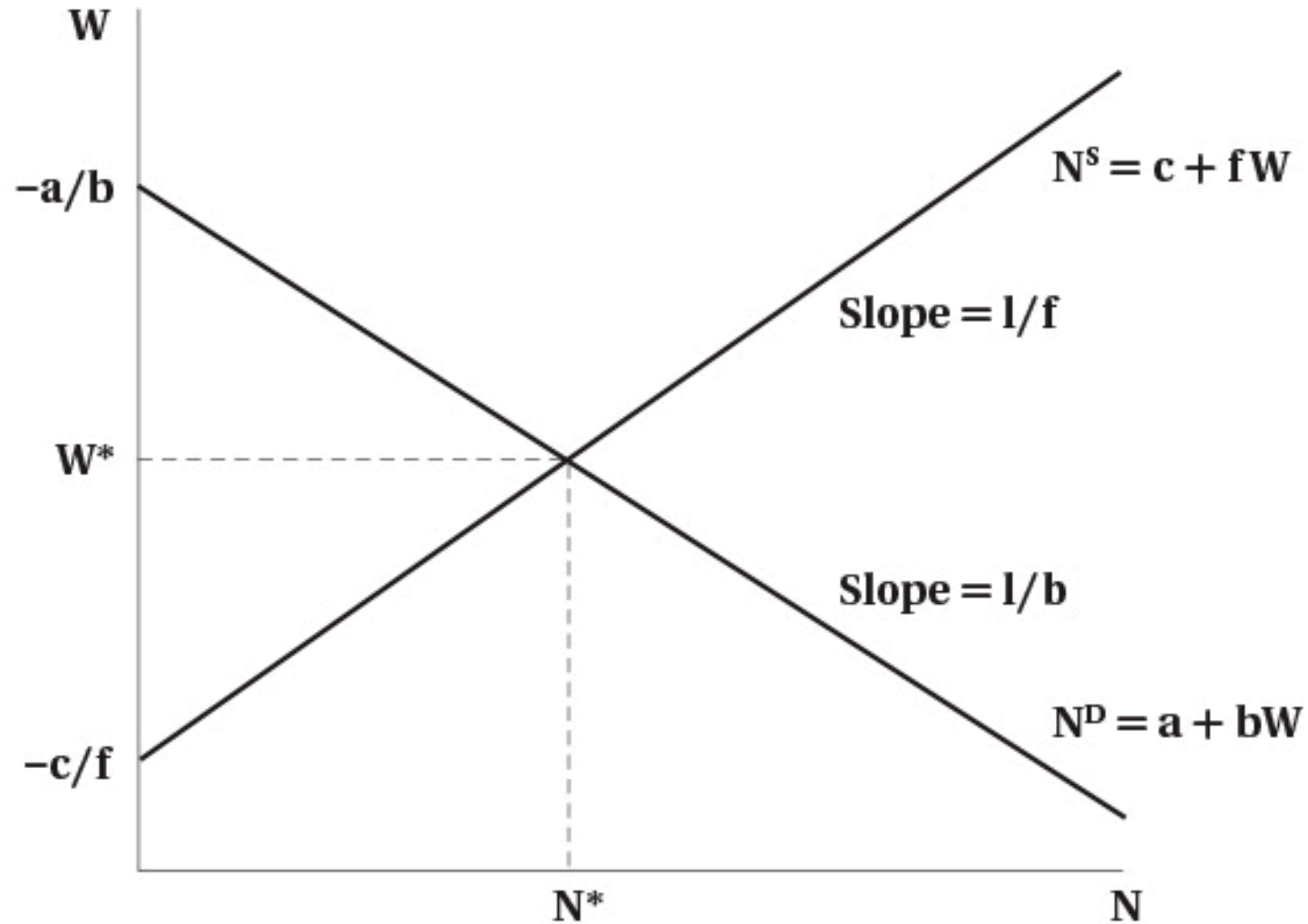
- Rearranging the supply function:

$$w = \frac{N^S - c}{f}$$

- Next slide plots these functions



# Demand and Supply Functions



**FIGURE 7.4** Linear Supply and Demand Functions

# Payroll Tax

- We can use this model to analyze various policies
- One common policy is a payroll tax on employers
  - Examples: Canada Pension Plan (CPP), Employment Insurance (EI)
- These are typically a percentage of wages paid by employers
- But we will analyze a fixed amount per hour for simplicity
  - Suppose the tax is  $T$  per hour
- This drives a “wedge” between the wage paid by employers and the wage received by workers in equilibrium

$$w_r = w_e - T$$

# Payroll Tax

- The demand function is:

$$w_e = \frac{N^D - a}{b}$$

- The supply function is:

$$w_r = \frac{N^S - c}{f}$$

# Payroll Tax

- Substitute into the equilibrium price equation and set quantity demanded equal to quantity supplied:

$$\frac{N^* - c}{f} = \frac{N^* - a}{b} - T$$

- Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af - bfT}{b - f}$$

# Payroll Tax

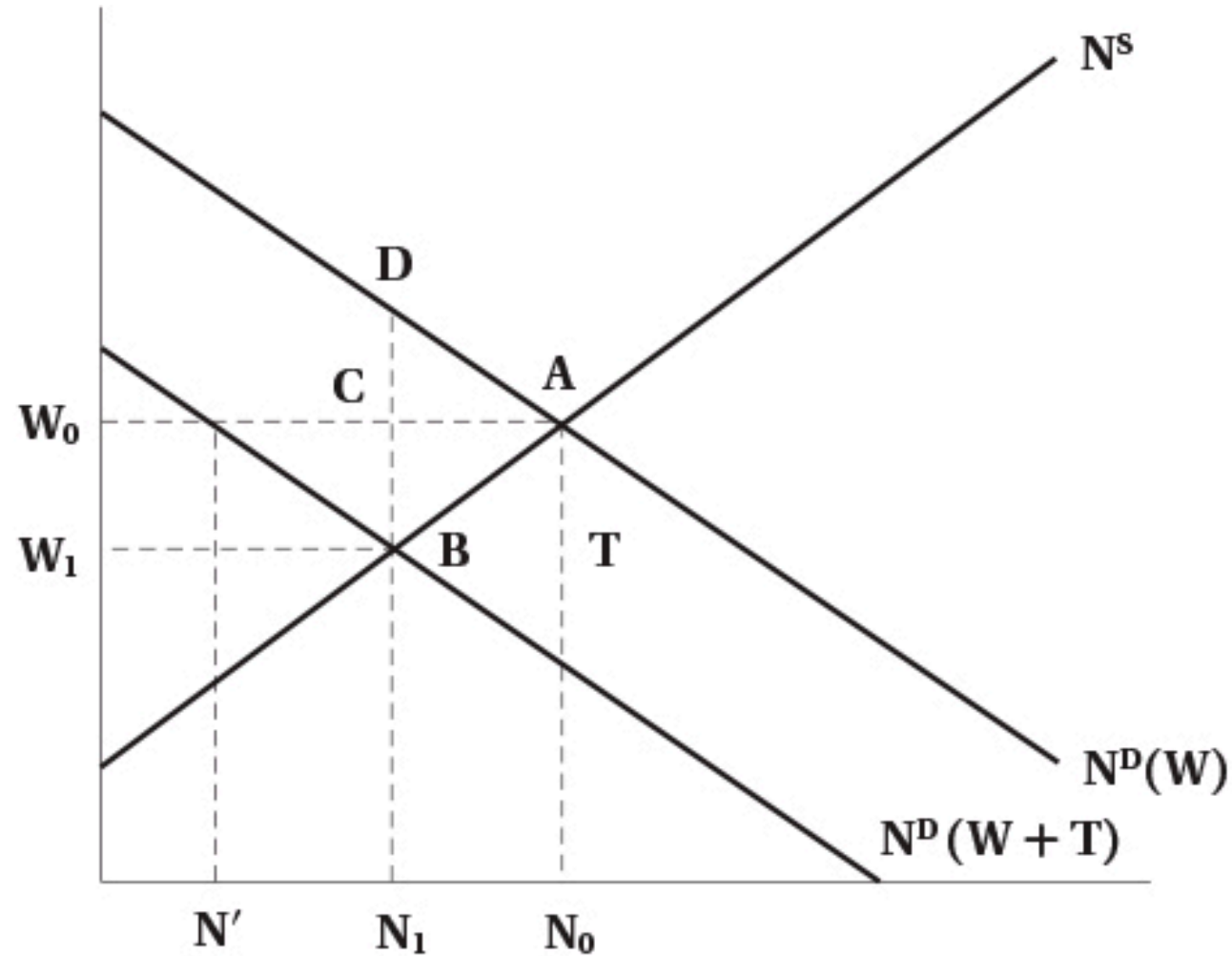
- The equilibrium wage received by workers is

$$w_r^* = \frac{c - a}{b - f} - \frac{b}{b - f} T$$

- The equilibrium wage paid by employers is

$$w_e^* = \frac{c - a}{b - f} - \frac{f}{b - f} T$$

# Payroll Tax



**FIGURE 7.5** The Effect of a Payroll Tax on Employment and Wages

# Payroll Tax

- The previous slide highlights the difference between the statutory and economic incidence of a tax
  - Statutory incidence: who is legally responsible for paying the tax
    - In this case, firms
  - Economic incidence: who actually bears the burden of the tax
    - Shared between firms and workers
    - Workers receive a lower wage than before
    - Firms pay a higher one
- Economic incidence depends on elasticities of supply and demand
  - More elastic supply means workers bear less of the tax burden
  - More elastic demand means firms bear less of the tax burden

# Evidence

- Theory predicts lower employment from a payroll tax
- Also predicts economic incidence may be shared between firms and workers
- Evidence based on empirical studies is
  - In the long run, workers bear most of the burden of the tax through lower wages
  - Employment effects are small



# Monopsony in the Labour Market

# Introduction

- Market power can be a feature on both the buying and selling side of markets
  - Monopoly: single seller with market power
  - Monopsony: single buyer with market power
- Typically we discuss monopoly in the goods market, but monopsony in the labour market
  - But you can have power in both buying and selling in both markets
- In this section we will consider monopsony in the labour market

# Introduction

- Some examples of monopsony in the labour market
  - Company/factory towns
  - Hospital labour markets
  - University professor labour markets
  - Public sector in small areas
- In all case there is a large buyer of labour
- This gives the buyer power to set wages
- Affects both wages and employment
  - To attract labour, it must raise wages

# Simple Monopsony

- In a monopsony, the firm faces the market supply curve for labour
  - Supply is upward sloping
  - Contrast to perfect competition where supply is horizontal
  - To hire more workers, the firm must pay a higher wage
- The firms marginal cost of labour is above the supply curve
  - Because to hire an additional worker, the firm must raise the wage for all workers
  - So MC is equal to cost of additional worker plus the increase in wage paid to existing workers
  - Analogous to how monopolists face MR below demand

# Simple Monopsony

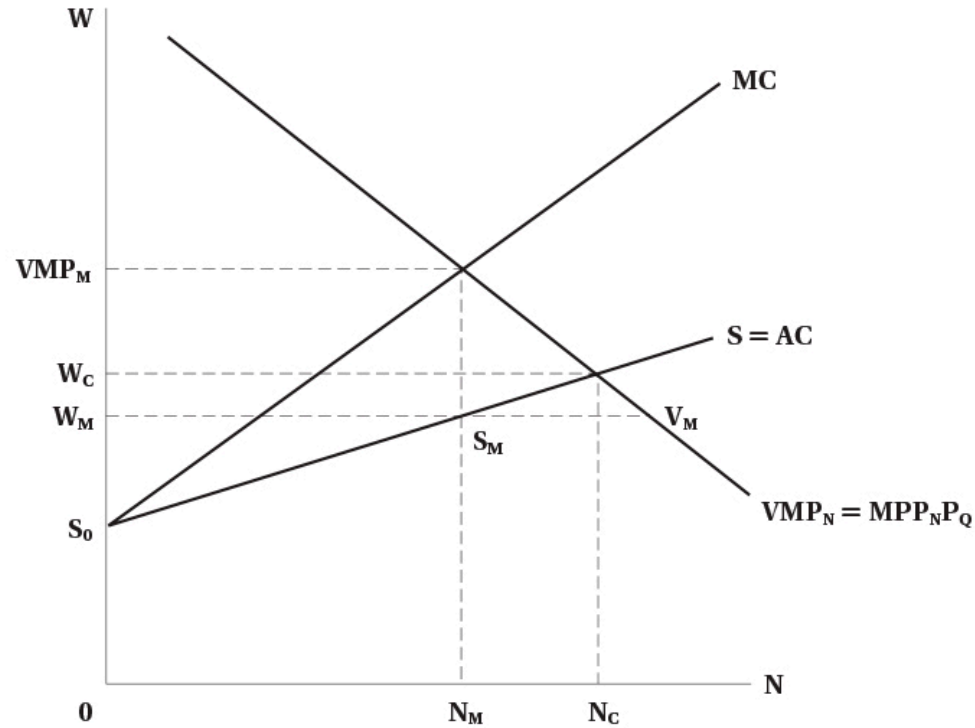


FIGURE 7.6 Monopsony

- Monopsony depicted to the left
- Firm demand is its value of marginal product
- Faces upward sloping supply curve
- MC is above supply
- Optimal labour is where  $MC = VMP$
- Wage comes from supply curve at that quantity of labour

# Simple Monopsony

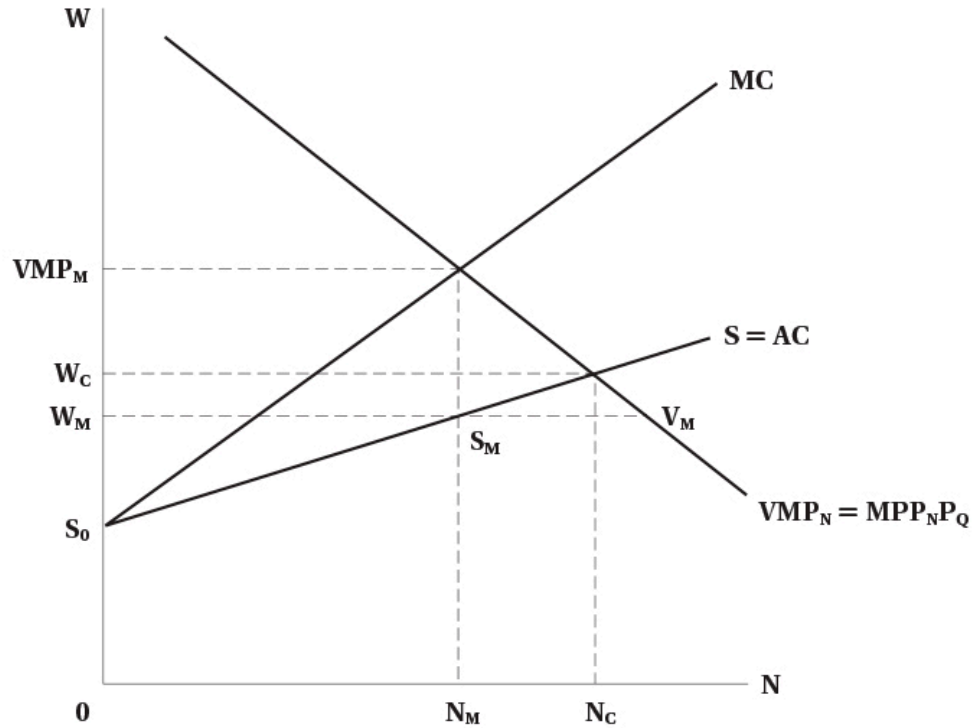


FIGURE 7.6 Monopsony

- For monopsony
  - Wage is below VMP
  - Wage is below competitive wage
  - Employment is below competitive employment
- Monopsony profit is rectangle  $(VMP_m - W_m) \times N_m$ 
  - Wage bill is  $W_m \times N_m$
  - Revenue contributed by workers is  $VMP_m \times N_m$

# Simple Monopsony

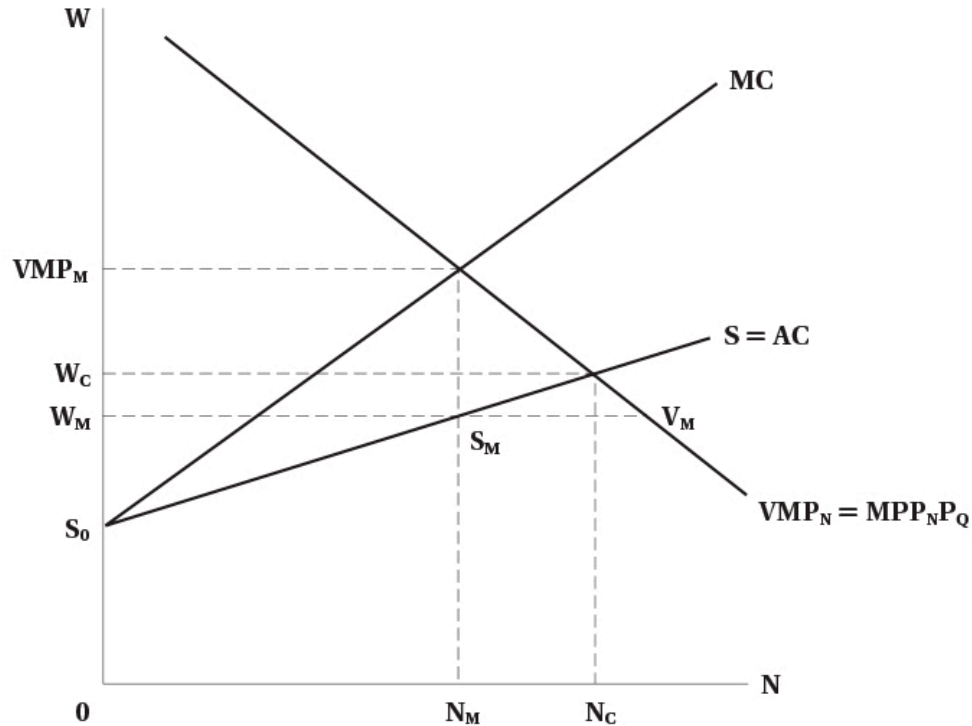


FIGURE 7.6 Monopsony

- Monopsonies also leave jobs vacant
  - Because they restrict employment to raise wages
  - This creates inefficiency in the labour market
- Vacancies are  $V_m - S_m$
- At wage  $W_m$ 
  - VMP is above wage
  - Firms would like to hire more -But to hire it must raise wages, so it is optimal to not hire more

# Characteristics of Monopsonies

- More prevalent in the short run
  - In the long run, workers can move to find other jobs
  - In the short run, frictions may prevent this
- Large relative to the labour market
  - Does not have to be a big firm, just large relative to the market
  - If the market is small, the firm does not have to be large to have power
  - Example: a restaurant in a seasonal tourist town
- Workers have specialized skills
  - May naturally be few employers for specialized labour
  - Like academia, sports teams



# Characteristics of Monopsonies

- Wages are lower than competitive equilibrium, but not necessarily low in an absolute sense
  - Example: sports players
  - Not paid low wages, but market power makes them lower than they would be otherwise

# Wage Differentiation

- In previous analysis, we assumed all workers are the same
  - Single supply curve for labour
- This leaves some surplus for workers
  - There are workers willing to work for less than the equilibrium wage
- Firm can capture that surplus by differentiating wages
  - Paying different workers different wages based on their reservation wage

# Wage Differentiation

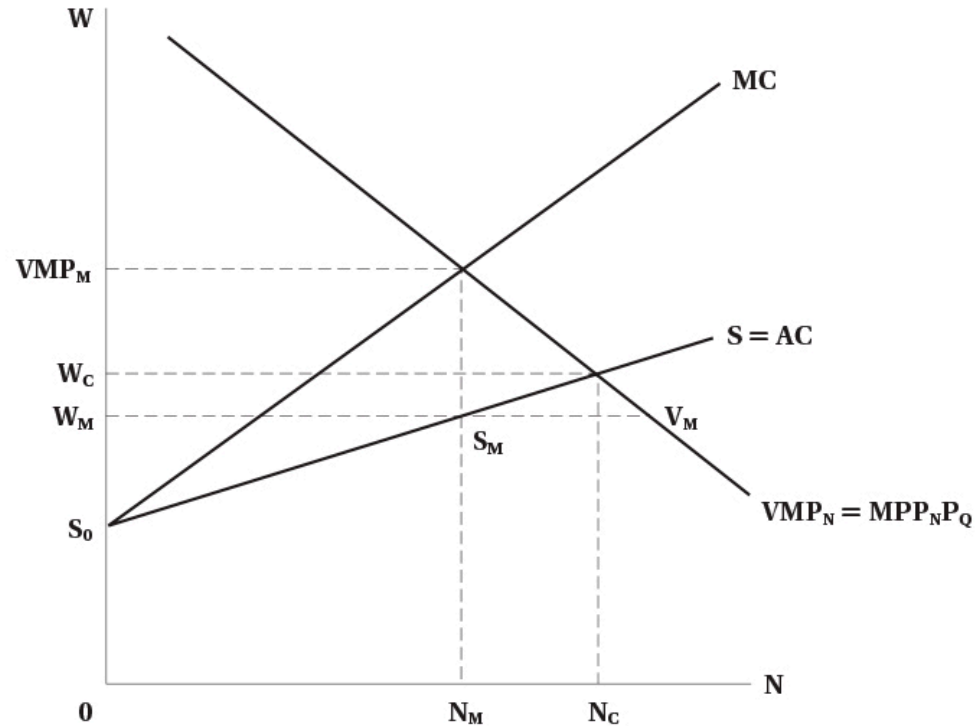


FIGURE 7.6 Monopsony

- If the firm can pay different wages
  - It can pay each worker their reservation wage
  - $MC$  curve and supply curve become equal to supply curve
- Firm will hire up to competitive level
  - Because it does not have to increase wage for all workers to hire more
- To do this, firms would try to make wages secret
  - Prevent workers from knowing what others are paid
  - Could do this with non-wage benefits for specific workers

# Evidence of Monopsonies

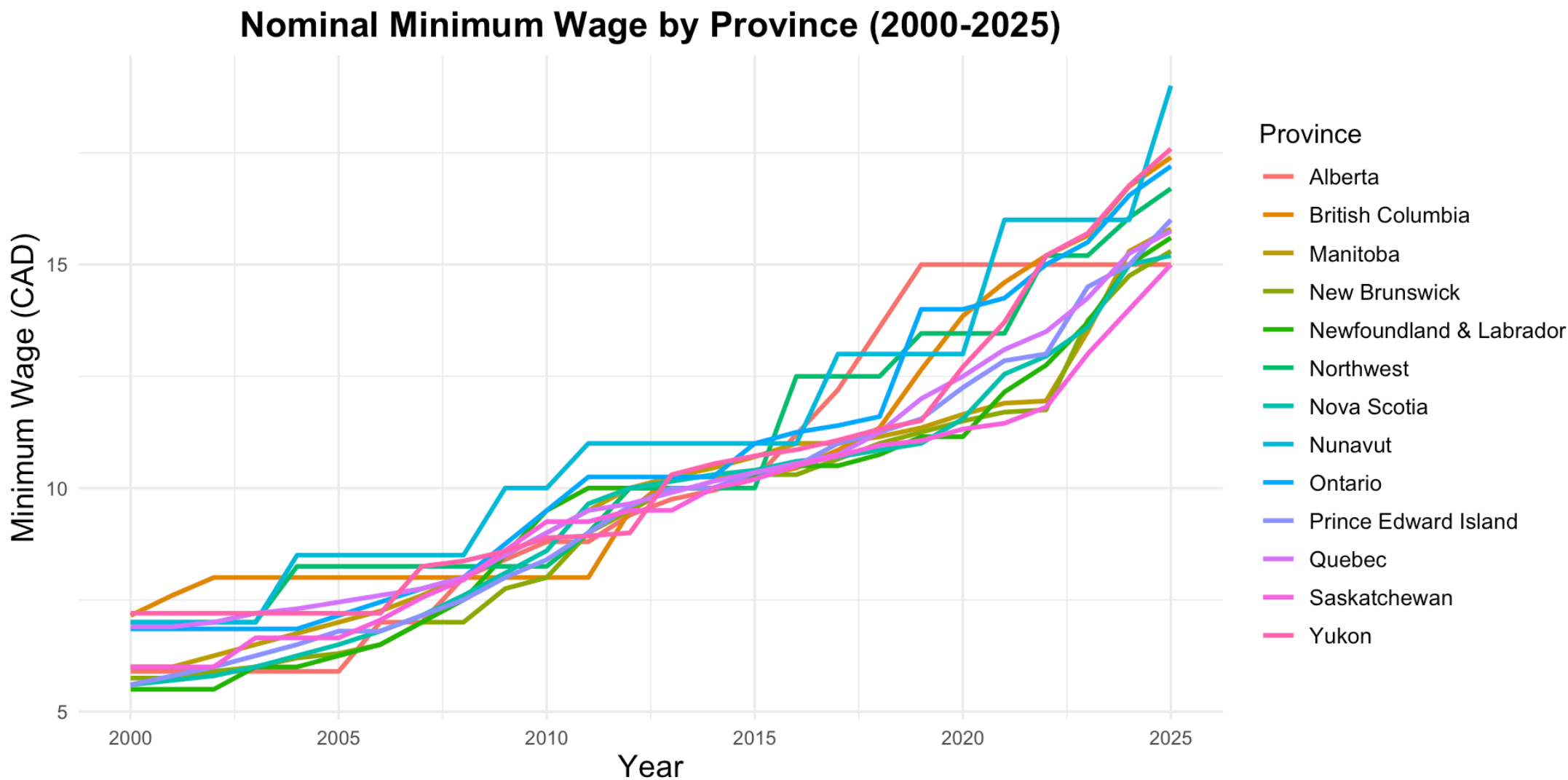
- Pro sports
  - Baseball: free agency increased wages significantly
    - Increases competition among teams
  - Hockey: not much evidence of monopsony power
- Academia and nursing
  - Some evidence of monopsony power
  - In Canada, a bargaining model may be more appropriate given these labour markets are unionized
- Some academics argue that monopsonies are relatively common
  - Firms can set wages only if they have some market power
  - Evidence that this is widespread

# Minimum Wages

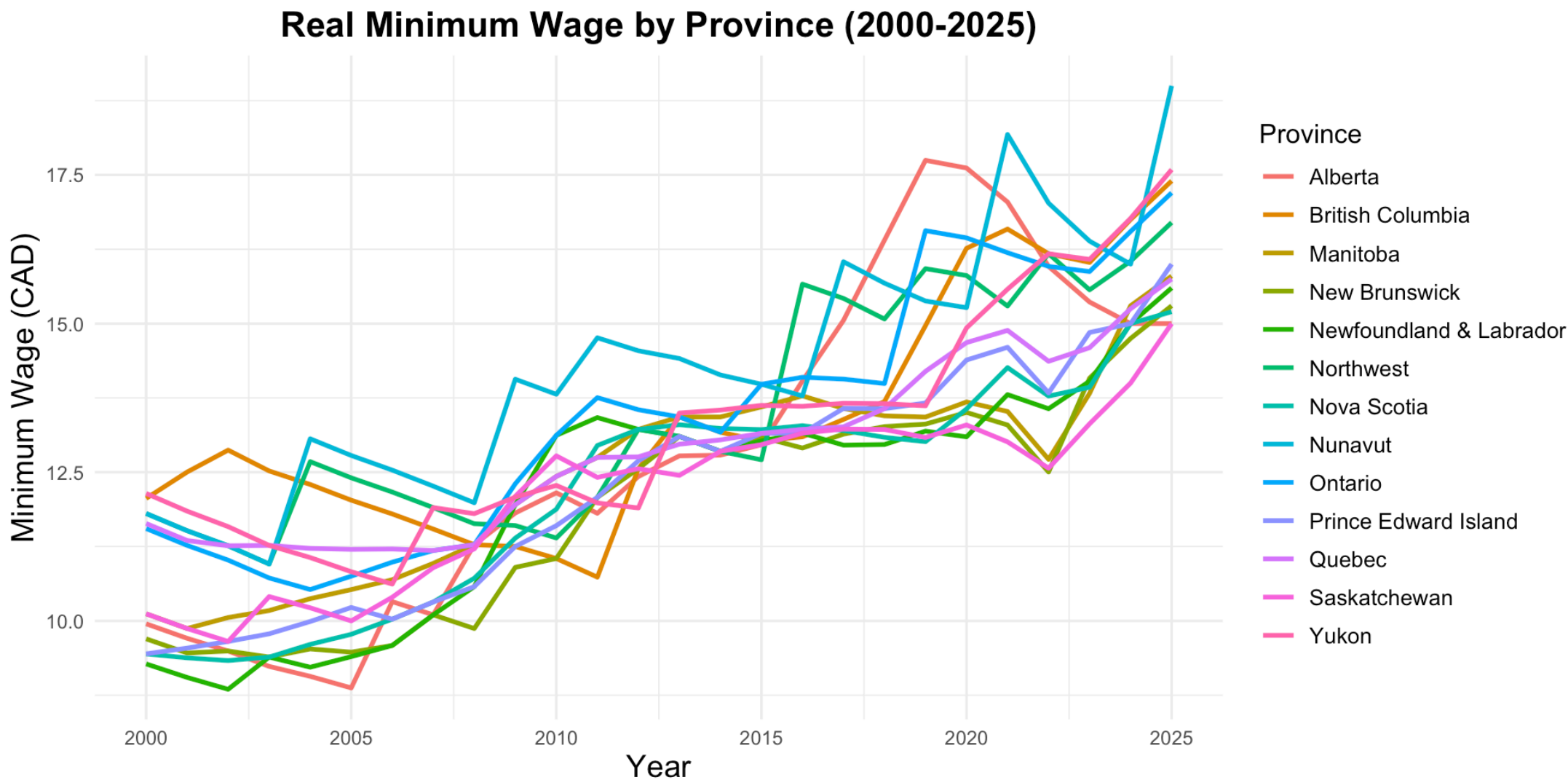
# Introduction

- Minimum wages are a common policy tool
  - Set a legal minimum wage that employers must pay workers
- Intended to raise wages of low-wage workers
  - Reduce poverty
  - Increase standard of living
- Standard economic theory predicts that minimum wages lead to unemployment
  - Because they set a wage above the equilibrium wage
  - Quantity of labour supplied exceeds quantity demanded
- But statistical evidence is much more mixed

# Minimum Wages by Province

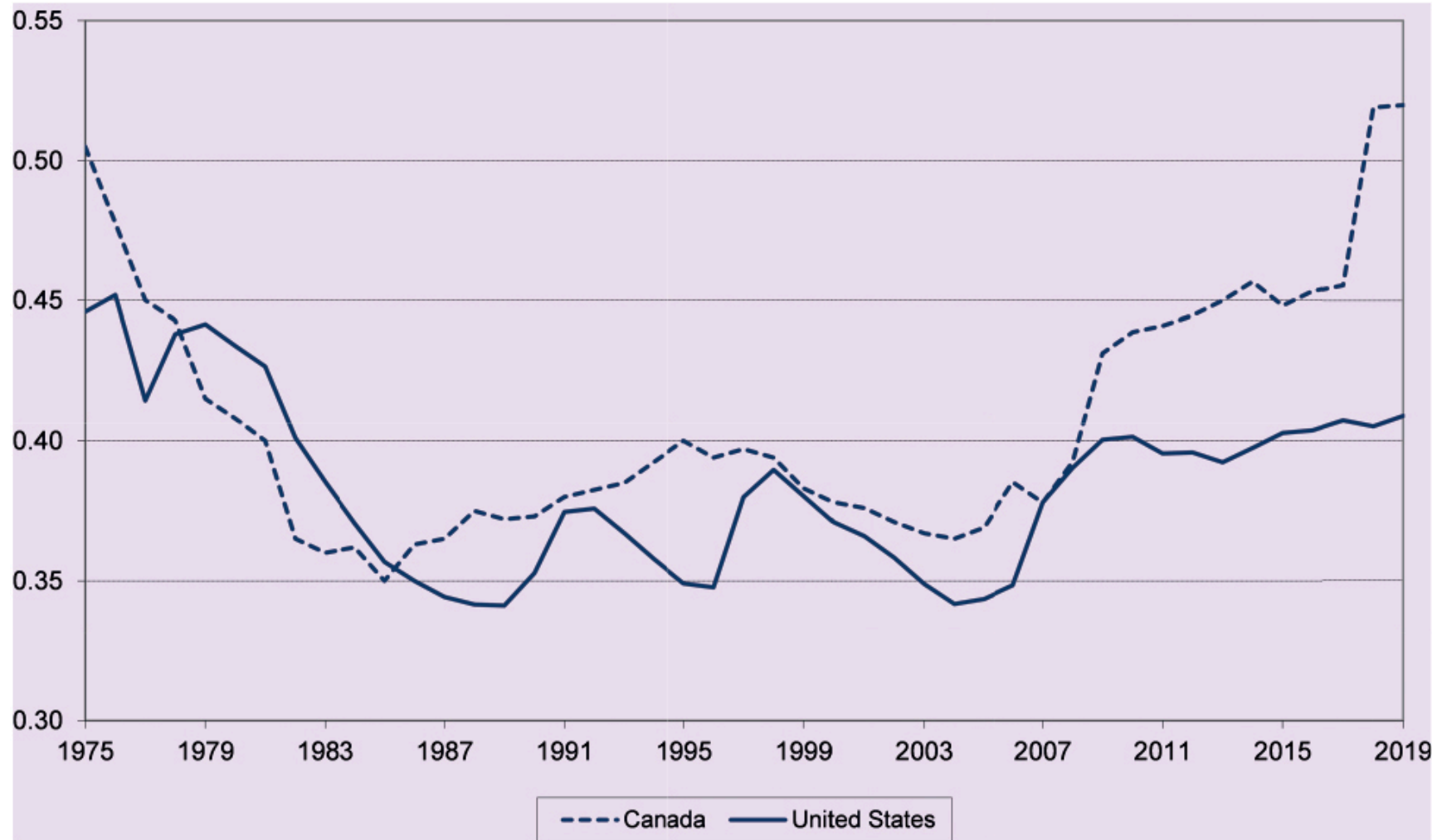


# Minimum Wages by Province





# Minimum Wages in Canada and USA



**FIGURE 7.7** The Ratio of Minimum Wages to Average Wages, Canada and the United States, 1975–2019

# Minimum Wages in Competitive Labour Market

- Suppose the labour market is competitive
- The market demand is downward sloping
- The market supply is upward sloping
- Minimum wages are introduced as a wage floor
  - Above the competitive equilibrium wage
- Creates excess supply of labour
  - Leads to unemployment

# Minimum Wages in Competitive Labour Market

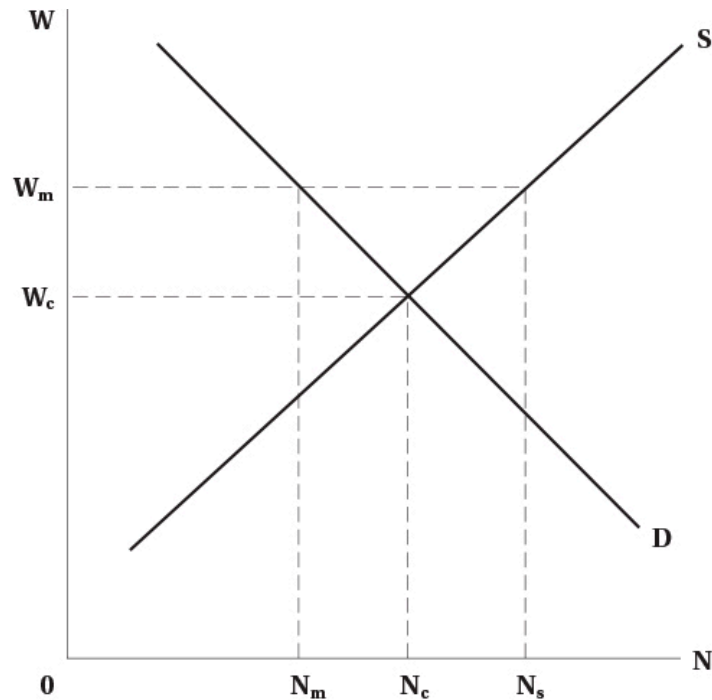


FIGURE 7.8 Minimum Wage

- Impact of minimum wage illustrated to the left
- Minimum wage is  $W_m$  above competitive level  $W_c$
- Creates excess supply of labour
- Leads to unemployment  $N_s - N_m$ 
  - People who would have been employed at competitive equilibrium  $N_c - N_m$
  - Others drawn into market by higher wage  $N_s - N_c$
-

# Minimum Wages in Competitive Labour Market

- Employment effects depend on elasticity of demand
  - More elastic demand means larger employment effects
- In real world, industries with minimum wage tend to have elastic demands
  - Lots of substitutes for low-wage labour
  - Labour costs are a large share of total costs
- Nevertheless, we will see that minimum wages may not affect employment much in practice

# Minimum Wages in Monopsony

- Effects of minimum wages are different in monopsonistic labour markets
- It is possible that minimum wages increase both wages and employment
  - Depending on where the minimum wage is set
- Graph on next slide illustrates this

# Minimum Wages in Monopsony

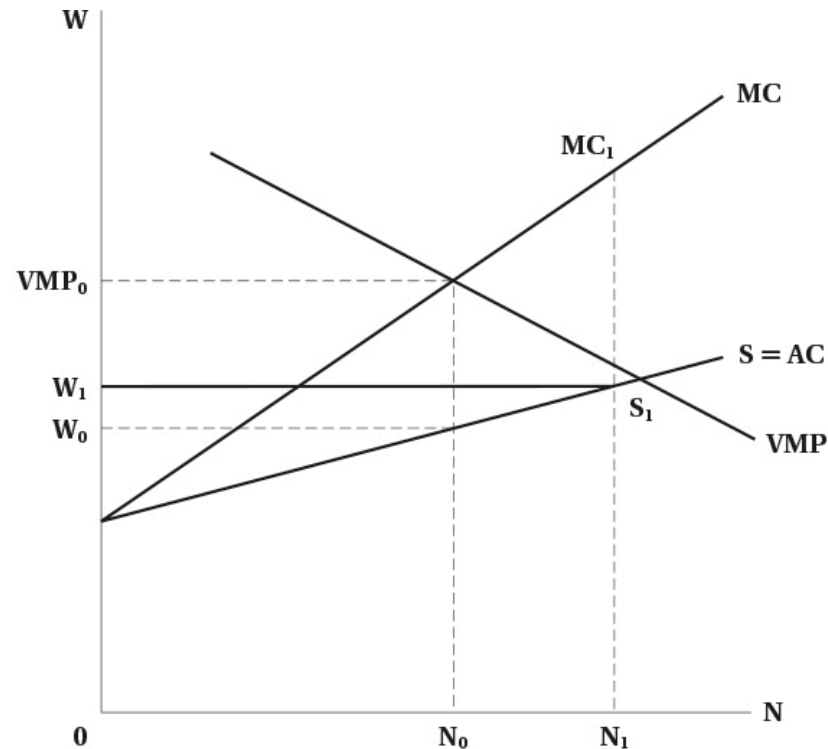


FIGURE 7.9 Monopsony and a Minimum Wage

- Graph shows standard monopsony
- Prior to minimum wage, wage is  $W_0$  and employment is  $N_0$
- Imagine minimum wage set above  $W_0$  but below competitive wage
- MC curve is minimum wage up to  $S_1$ 
  - Then jumps up to previous MC curve
- Optimize where  $MC = VMP$ 
  - Employment increases to  $N_1$

# Minimum Wages in Monopsony

- Why would minimum wages cause employment to increase?
  - Previously, firm had to increase wages to attract more labour
  - With minimum wage, it does not
    - Everyone is paid the same wage
    - Wage increases no longer inhibit employment growth
  - Effectively, minimum wage lowers marginal cost
- This may be partly why we do not see large employment effects of minimum wages
- Note that monopsonists do not like minimum wages
  - They reduce their market power
  - Increases costs, reduces profits

# Offsetting Factors for Minimum Wage

- As usual, in models we hold all else equal
- But in reality, many things are changing at once
  - Minimum wage could be accompanied by changes in demand
    - For example, fiscal stimulus increasing demand for goods and labour
  - Minimum wage could lead to changes in technology
    - May change demand for labour
  - Firms might choose to absorb minimum wage costs



# Evidence

- Minimum wages are one of the most researched areas in labour economics
- Large amounts of evidence goes back at least 50 years
- Early studies looked at time trends in employment and minimum wages
  - Subject to a lot of bias
  - Things often trend together over time without direct causation
- Research in early 1990s used better methods
  - Natural experiments
  - Difference-in-differences

# Evidence

- Most prominent study is Card and Krueger (1994)
  - Looked at fast food restaurants in New Jersey and Pennsylvania
  - New Jersey increased minimum wage, Pennsylvania did not
  - Used difference-in-differences to estimate employment effects
  - Found no negative employment effects from minimum wage increase
- Has been reevaluated several times
  - Conclusion remains that employment effects are small or non-existent

# Evidence

- In Canada, minimum wages increase youth unemployment
  - Studies looked at minimum wage changes across provinces in 1990s
- Some Canadian studies look deeper at labour market dynamics
  - In any month, there is a lot of turnover in the labour market
  - Each month roughly 6% of employed people separate, 8% of non-employed people hired
  - Evidence shows minimum wages reduce both hiring and separations
  - Leads to no effect overall

# Evidence

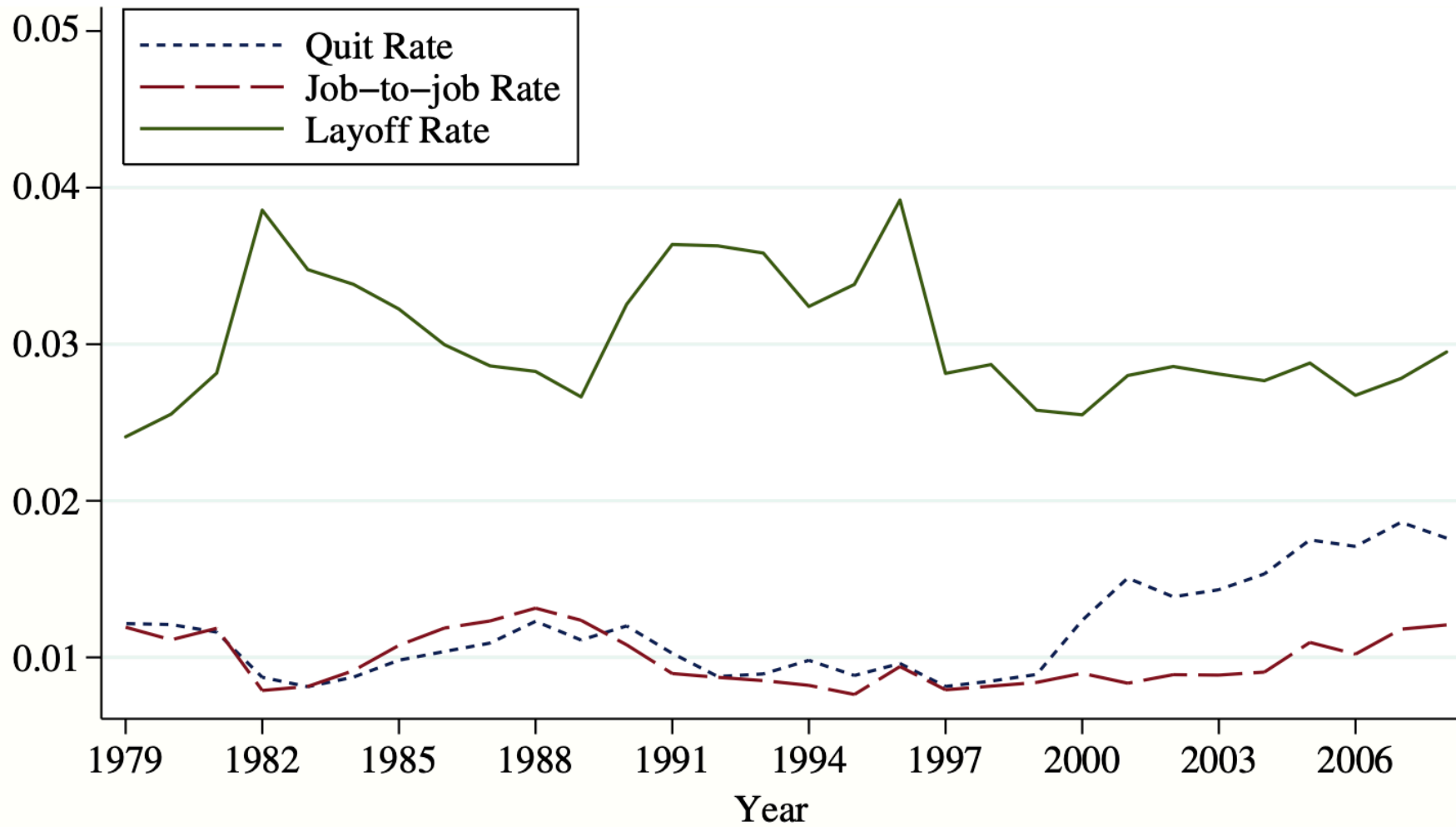


Fig. 4. *Layoff and Quit Rates, Low Skilled, 1979–2008*

# Evidence

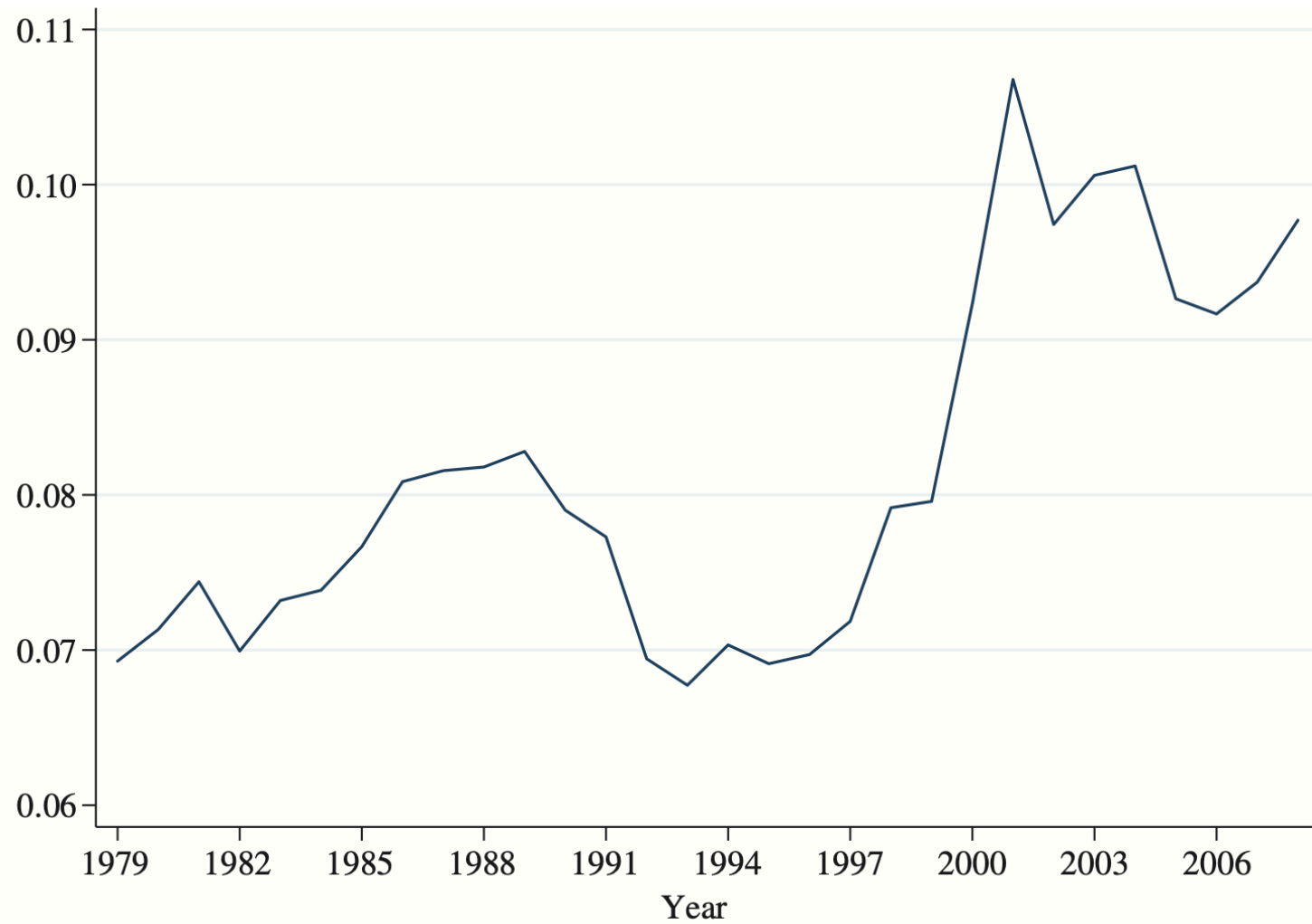


Fig. 5. *Hiring Rates, Low Skilled, 1979–2008*

# Evidence

- Also recall that one point of minimum wage is to reduce poverty
- Evidence shows they are generally not successful
  - Both Canada and US studies show no link
- Why do they not work?
  - Many poor individuals do not work
  - Many people on minimum wages are not poor
    - Youth living in high income families

# Summary

# Summary

- We analyzed how wages and employment are determined in a single labour market
- Market power in the goods market reduces both wages and employment
- Market power in the labour market reduces wages and employment
- Minimum wages may reduce employment in competitive markets
- Minimum wages may increase employment in monopsonistic markets
- Empirical evidence shows minimum wages have small effects on employment